



This data validation is critical to ensuring accurate analysis downstream. You could then build interactive dashboards that pull data through OFFSET and INDIRECT functions to visualise metrics and trends.

This allows you to easily compare performance across plants, processes, and products and identify lag areas for deeper analysis using Excel. For example, a plant with lower OEE could be traced back to high downtimes, which could be mapped to specific unscheduled breakdowns and loss categories.

Optimizer Model for Improvement Roadmap

You can develop an optimisation model for the final recommendations to identify and prioritise projects to improve operational metrics.

This model used inputs and calculations across multiple worksheets, such as:

Loss analysis worksheet mapping losses to underlying root causes

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Solution worksheet listing potential projects, benefits and costs

Know More...

Constraint worksheet capturing implementation considerations

Know More...

Master optimiser worksheet evaluating all possible portfolios

Know More...

Using a combination of Solver for optimisation and scenario manager, you could identify the project roadmap to maximise metrics like OEE improvement within budgets and other constraints.

The ability to quickly build custom optimisation models using Excel functions and features and powerful visual calculative capabilities made this process extremely efficient and collaborative with the client team.

This case highlights how spreadsheets can streamline processes like:

- ✔ Standardising data collection and collation
- ✔ Validating data quality through integrated checks
- ✔ Developing visual dashboards for performance tracking checks
- ✔ Building custom optimisation models with ease
- ✔ Collaborating seamlessly to develop insightful recommendations